KS3: INDUSTRIALISATION, EMPIRE, AND POWER (1750–1900)

COURSE TEXTBOOK & TEXTBOOK

Enquiry: *Industrialisation, Empire, and Power: How did 19th-century Britain transform at home and abroad?*

Contents

L1: What powered the Industrial Revolution, and how did a Fareham ironmaster change the world?

LEARNING OBJECTIVES

- ☐ Understand the difference between the domestic system and the factory system.
- ☐ Explain the significance of Henry Cort's puddling and rolling processes.
- ☐ Evaluate the importance of location versus innovation in Cort's success.

Do Now Activity

1. What is meant by the term 'Divine Right of Kings'?

2. Who became King after the English Civil War ended and Oliver Cromwell died?

3. Why did the Gunpowder Plot of 1605 fail?

4. What was the significance of the Magna Carta signed in 1215?

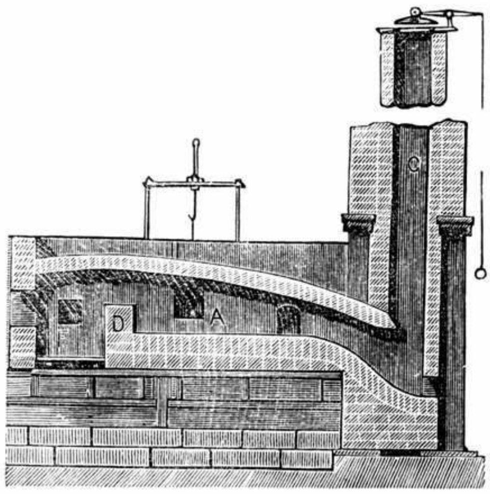
5. Why did Henry VIII break away from the Catholic Church?

Vocabulary Check

Write a short paragraph using at least FOUR of the vocabulary words below correctly.

Industrial Revolution | Pig Iron | Wrought Iron | Puddling Process | Rolling Mill

Source A: The Puddling Furnace



A historical cross-section diagram of a puddling furnace, the revolutionary process Henry Cort pioneered at his Funtley ironworks in Hampshire.

Before the 1700s, Britain was an agricultural country. Most goods were made by hand in people's homes (the "domestic system"), and the only machines that existed were powered by natural, unreliable sources like wind, fast-flowing rivers, or the muscle power of horses. However, beneath Britain's soil lay massive reserves of **coal**—a fuel that burned much hotter than wood. In 1712, Thomas Newcomen invented the first practical steam engine, which used burning coal to pump water out of deep mines. Later, in 1769, an engineer named **James Watt** drastically improved the design, making the steam engine much more efficient and powerful. Watt's steam engine changed everything: it meant that massive factories no longer needed to be built next to rivers. They could be built anywhere, powered entirely by coal and steam. This shift from farming and hand-crafting to mass factory production driven by steam power is known as the **Industrial Revolution**. However, to build these enormous new steam engines, factories, railways, and bridges, Britain desperately needed one crucial material: **iron**.

Q1. Why was James Watt's improvement of the steam engine so significant for the location of factories?

In the early 1700s, British iron was of terrible quality. It was known as "pig iron"—it was brittle and snapped easily because it was full of impurities (carbon). If the Royal Navy wanted strong, flexible iron (known as "wrought iron") for its ship masts and cannons, it had to buy it from Sweden or Russia at a massive cost.

The man who solved this national crisis lived right here in Hampshire. Henry Cort was an ambitious navy agent who realized he could make a fortune if he could figure out how to mass-produce high-quality British iron. In 1775, he took over the **Funtley Ironworks** (just north of Fareham) because it was ideally located to supply the massive Royal Navy dockyards in nearby Portsmouth. At Funtley, Cort invented two revolutionary processes in 1783 and 1784: 1. **The Puddling Process:** He invented a new type of furnace where molten iron was stirred (or "puddled") with long iron rods to burn off carbon impurities. 2. **The Rolling Mill:** Instead of hammering the iron into shape by hand, Cort passed the red-hot iron through heavy, grooved rollers to squeeze out the last impurities and shape it into rails and bars.

KNOWLEDGE CHECK (Q5)

What was the primary purpose of the 'Puddling Process' invented by Cort?

- ☐ A. To hammer the iron into shape by hand.
- ☐ B. To stir molten iron and burn off carbon impurities.
- ☐ C. To make iron brittle and easy to snap.
- ☐ D. To import iron from Sweden more efficiently.

Q2. Why did Henry Cort decide to take over the Funtley Ironworks in 1775?

Q3. What two major inventions did Henry Cort patent at Funtley in the 1780s?

Q4. Study Source A. How does the physical design of the puddling furnace show that Cort was trying to solve the problem of 'pig iron' impurities?

Cort's inventions at Funtley changed the world. They allowed British iron production to multiply by 400% over the next twenty years, earning Cort the nickname "the father of the iron trade." The rails for the new train networks and the iron for the ships that built the British Empire were made using Cort's methods. Tragically, Cort did not die a wealthy hero. His business partner, Adam Jellicoe, had secretly stolen the money used to fund the Funtley Ironworks from the Royal Navy. When Jellicoe died, the government seized Cort's patents and ruined him. Cort died bankrupt and in poverty, but his local ingenuity powered a global empire.

Q6. Why did Cort die in poverty despite his world-changing inventions?

Q7. Why were Henry Cort's inventions so vital to the growth of the British Empire?

When assessing why the Funtley Ironworks was so successful in kickstarting the Industrial Revolution's iron industry, historians debate two main factors: **1. Geographical Luck (Location):** Funtley was situated just a few miles from the Portsmouth Royal Navy dockyards. In the 18th century, before the invention of trains, transporting heavy iron overland was incredibly expensive. Being located right next door to the largest industrial buyer of iron in the world (the Royal Navy) meant Cort had a massive, guaranteed market for his goods. Geography provided the economic opportunity. **2. Technological Genius (Innovation):** However, location alone wasn't enough. The Royal Navy desperately needed *wrought* iron (flexible and strong) for its ships and cannons, not the brittle British *pig iron* that snapped easily. Without Cort's brilliant inventions—the Puddling Furnace to burn off impurities and the Rolling Mill to speed up shaping—he would have had nothing useful to sell to the Navy. Innovation provided the solution to a national crisis. Historians must weigh these two factors to decide which was *more* important to his success.

Q8. Write a structured paragraph answering the following: "To what extent was the proximity to the Portsmouth dockyards the most important factor in the success of the Funtley Ironworks?"

Q9. Chronology Challenge: Place the following in the correct causal order and write one sentence explaining how each led to the next: Watt's Steam Engine, The Domestic System, Cort's Puddling Furnace.

L2: Was industrial work progress or punishment?

LEARNING OBJECTIVES

- ☐ Understand the working conditions of the early Industrial Revolution with historical nuance.
- ☐ Evaluate primary sources regarding child labour.
- ☐ Analyse the 'Optimist vs. Pessimist' historical debate.

Do Now Activity

1. **What two major iron-making processes did Henry Cort patent at the Funtley Ironworks in the 1780s?**
2. **Why did Henry Cort specifically choose to locate his ironworks near Fareham?**
3. **What tragic event caused Henry Cort to lose his patents and die in poverty?**
4. **What was the primary cause of the English Civil War?**
5. **What was the result of the Spanish Armada in 1588?**

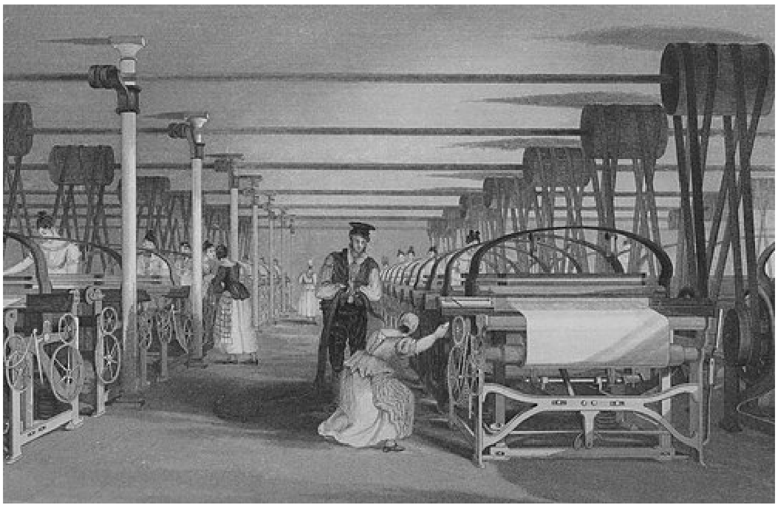
Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Fareham Reds | Pug Mill | Primary Source | Historiography | Factory Acts

Your Sentence:

Source A: Powerloom Weaving in 1835



A 19th-century engraving depicting child labour in an early industrial textile mill.

Before the 1700s, most goods were manufactured by hand in people's homes, known as the 'domestic system'. It is a common myth that this was a relaxed, golden age; in reality, working at home was backbreaking, poorly paid, and children worked long hours weaving or spinning. However, the transition to the 'factory system' completely transformed the scale and danger of British life. As steam engines became more powerful, massive factories were built, causing rapid urbanisation. On a national level, this was incredible progress: Britain became the 'workshop of the world'. But for the workers, the reality was a brutal shock. The machines now dictated the pace of life. Shifts lasted 12 to 14 hours, six days a week. The noise was deafening, the air was polluted, and heavy machinery lacked safety guards, leading to horrific, unprecedented injuries.

Q1. What was the main difference between the child labour used in the 'domestic system' compared to the new 'factory system'?

Factory and mine owners actively sought out child workers. Children were cheap (paid 10% of an adult's wage), easier to discipline, and small enough to crawl under moving machinery. In 1832, Parliament sent Michael Sadler to investigate these conditions. His official report shocked the nation. One textile worker, Matthew Crabtree, was interviewed: 'I began work at eight years old... our common time was from six in the morning till half-past eight at night... we were frequently strapped [beaten] if we were too late or started to fall asleep.' The economic success of the early Industrial Revolution was largely built on the broken backs of these children.

Q2. How useful is the 1832 Sadler Report for a historian studying the reality of child labour?

The Industrial Revolution didn't just happen in northern cotton mills; it radically changed the landscape of Hampshire. As cities like London exploded in size, they desperately needed building materials. Fareham sat on a geological goldmine: deep deposits of high-quality clay. During the 19th century, the Funtley Clay Pits and Fareham Brickfields expanded massively. The unique clay produced a brilliant, vibrant red brick famously known as 'Fareham Reds'. These bricks were so highly prized by Victorian architects that they were transported by the newly built railways to construct prestigious buildings in the British Empire, including the Royal Albert Hall in London. Fareham's local mud was literally building the modern world.

Q3. Sort the following factors into the correct chronological order to explain the boom in Fareham's economy: A) Railways are built to transport heavy goods, B) London's population explodes, C) Fareham Reds are used to build the Royal Albert Hall, D) Massive demand for building materials is created.

While the Fareham Reds brought great wealth to the brickyard owners, the reality for the workers was grueling. In the Funtley brickfields, whole families worked together. Young boys and girls were employed as 'pug boys', driving the horses that turned the pug mills to crush the heavy clay. Others worked as 'carriers', lugging wet clay, or as 'turners', walking up and down long outdoor drying hacks to manually flip thousands of heavy bricks so they dried evenly. We know this wasn't just happening in the north of England; if a historian looks at the 1881 Census returns for the Fareham area, they will find numerous local children, some as young as 10, officially listed with occupations like 'brickmaker's labourer' or 'pug boy', proving the local economy relied heavily on child exploitation.

Q4. Why might a historian need to look at BOTH a national interview (like the Sadler Report) AND a local document (like the 1881 Fareham Census) to fully understand child labour?

As the 19th century progressed, the horrific human cost disgusted the public, leading Parliament to pass 'Factory Acts' to restrict child labour and mandate schooling. But historians still fiercely debate the overall impact of the Industrial Revolution. 'Pessimist' historians (like E.P. Thompson) argue that industrialisation was a catastrophe that punished the working class, destroying their traditional way of life and exploiting children for decades. Conversely, 'Optimist' historians (like R.M. Hartwell) argue that while the initial transition was harsh, it ultimately represented massive progress; it created the wealth, medical advancements, and eventual higher wages that dramatically increased the life expectancy and standard of living for everyone in the 20th century.

Q5. Write a structured paragraph answering: 'To what extent was the Industrial Revolution a

punishment rather than progress for the working classes?'

L3: Did industrialisation make British towns unlivable?

LEARNING OBJECTIVES

- ☐ Analyse the contrast between imperial wealth and working-class squalor.
- ☐ Evaluate the usefulness of contrasting primary sources.
- ☐ Understand the catalysts for public health reform.

Do Now Activity

1. What are 'Fareham Reds'?

2. Name one highly dangerous or physically demanding job a child might perform in a 19th-century textile mill or brickfield.

3. Why did factory owners actively prefer to hire children rather than adults?

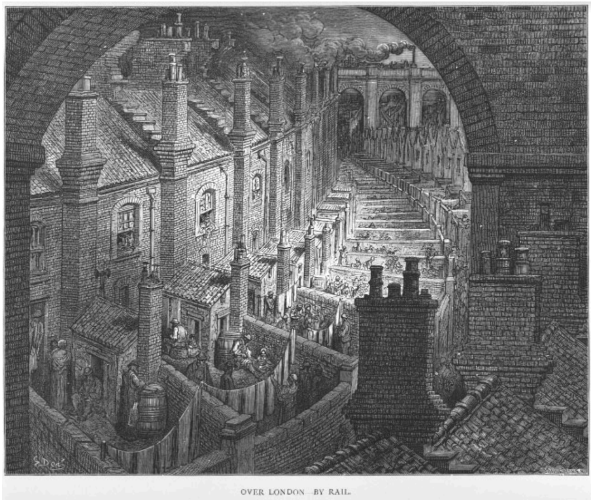
4. Explain how Henry Cort's inventions at the Funtley Ironworks helped power the Industrial Revolution globally.

5. Why do 'Optimist' and 'Pessimist' historians disagree about the impact of the Industrial Revolution on working-class families?

Vocabulary Check

Write a clear definition and a historically accurate sentence for the term: ***Urbanisation***

Visual Hook: Over London by Rail (1872)



An engraving by Gustave Doré depicting the overcrowded and polluted back-to-back housing of Victorian London, published in 1872.

Source B: The Unregulated Capitalism / Corporate Perspective

> *"The demand for the Fareham bricks remains unprecedentedly high, owing to their great durability under heavy pressure. Contracts have been secured for the building of the vast railway arches, the expanding dockyards at Portsmouth, and prestigious public buildings in the capital. To keep pace with the building of these modern towns, the clay pits must work without interruption, and cottages for the labourers must be raised instantly adjacent to the works to secure the utmost efficiency of labour." — Adapted from 19th-Century Southern Industrial and Brick-making Directory Records*

During the 19th century, Britain's population boomed and millions flooded into cities. Construction exploded. To understand the scale of this, historians look at sources like Source B, adapted from an industrial directory for Victorian contractors (like Joseph Bull & Sons). This shows the 'façade' of the Empire: wealthy exteriors built by local Hampshire mud. But behind these glorious public buildings lay a very different reality for the working classes.

Q1. Conceptual Analysis: Based on Source B, what was the primary concern of the Victorian contractors when building 'cottages' for their workers?

Source A: The Official, Top-Down Government Data

> *"The charcoal-burners, the brick-makers, and the miners are all exposed to severe atmospheric changes and filth... In the town districts, the crowded back-to-back dwellings are built with no ventilation, and the refuse of the houses is thrown into open streets. The annual loss of life from filth and bad ventilation is greater than the loss from death or wounds in any wars in which the country has been engaged in modern times." — Edwin Chadwick, Report on the Sanitary Conditions of the Labouring Population of Great Britain, 1842*

To house the exploding population, landlords packed workers into tiny 'back-to-back' houses sharing three walls, offering zero ventilation. Entire families lived in a single room with no plumbing, sharing a single outdoor toilet over a deep cesspit. We know about this squalor from official investigations, such as Source A, an 1842 government report by reformer Edwin Chadwick.

Q2. Language Analysis: Look at Source A. Why do you think Edwin Chadwick chose to compare the deaths in the slums to the deaths in a 'modern war'?

Source C: The Bottom-Up, Raw Human Perspective

> *"We live in muck and filth. We aint got no privies, no dust bins, no drains, no water-splies, and no sewers in the whole place... We are living like pigs, and it aint fair. We hope you will print this to let the great people know how we are left to die of the cholera. We are your regular readers, and we pray you to help us." — Letter to the Editor of The Times, signed by 54 residents of London slums, 1849*

While Chadwick's report provided official statistics, historians also need the raw, human perspective to fully understand the crisis. Source C is an authentic letter sent to The Times newspaper in 1849 by 54 desperate residents of a London slum during a deadly disease outbreak.

[[SRC_MARKER:L2_Task_2_0]] **Q3. Source Comparison: How does Source C provide a different historical perspective to Source B?**

The ultimate consequence of this squalor was Cholera, a deadly waterborne disease. Initially, Victorian doctors believed in the 'Miasma Theory' (that bad smells caused disease), so they burned barrels of tar. In 1854, Dr. John Snow proved cholera was in the water. Despite this, the government maintained a 'laissez-faire' (do nothing) policy because improving sanitation was expensive. This changed during the blazing hot summer of 1858. The 'Great Stink' of raw sewage drying in the River Thames became so unbearable that politicians in Parliament had to soak their curtains in chemicals to mask the stench. The physical reality of the smell finally forced the government to abandon laissez-faire, funding engineer Joseph Bazalgette millions to build a massive underground brick sewer system.

[[SRC_MARKER:L2_Task_3_0]] **Q4. Turning Point: Why did it take the 'Great Stink' for the government to finally abandon its 'laissez-faire' policy?**

Following the construction of the sewers, Parliament passed the 1875 Public Health Act, finally forcing local councils to provide clean water and drainage. Today, historians debate the legacy of this era. 'Pessimist' historians argue that industrialisation was a catastrophe that subjected a whole generation to unlivable, deadly slums, pointing to the staggering cholera death tolls of the 1830s and 40s. Conversely, 'Optimist' historians argue that this crisis was a necessary growing pain. The rapid urbanisation forced the government to modernize and abandon laissez-faire; the immense wealth generated by the factories eventually paid for engineering marvels like Bazalgette's sewers and the laws that created the safe, sanitary cities we live in today.

[[SRC_MARKER:L2_Task_4_0]] **Q5. Extended Writing: Write a structured paragraph answering: 'To what extent did industrialisation make British towns unlivable?'**

L4: How was the British Empire built and sustained?

LEARNING OBJECTIVES

- ☐ Understand the transition of imperial rule from private mercantilism to direct state control.
- ☐ Evaluate how domestic industrial complexes sustained global naval supremacy.
- ☐ Deconstruct primary source provenance to determine historical utility.

Do Now Activity

1. Why did Victorian builders heavily rely on 'Fareham Red' bricks during the rapid urbanisation of the 19th century?

2. How did the 1842 Chadwick Report challenge the traditional government policy of 'laissez-faire'?

3. What scientific breakthrough did Dr. John Snow achieve during the 1854 cholera outbreak in Soho?

Vocabulary Check

Write a short paragraph using at least FOUR of the vocabulary words below correctly.

Mercantilism | British Raj | Two-Power Standard | Ironclad | Captive Market | East India Company (EIC) | Sepoy Mutiny | Naval Supremacy

HMS Warrior at Portsmouth

Source

Launched in 1860, HMS Warrior was the world's first iron-hulled, steam-powered ironclad warship, representing the absolute peak of British industrial and naval supremacy.

The British Empire of the 19th century was the largest empire in human history, yet its grandest conquest did not begin with state military planning. Instead, it was executed by a private, profit-driven corporation: the East India Company (EIC). Founded in 1600 to control trade in precious spices, silks, indigo, and tea, the EIC transformed into an aggressive imperial powerhouse. To secure its commercial monopolies, the Company recruited its own private mercenary army, exploiting political divisions between local rulers. By systematically weaponizing its financial might, the EIC turned the Indian subcontinent into a captive market. This drove a ruthless economic cycle: India's rich fields produced raw cotton, which was shipped back to feed the brutal factories studied in Lesson 2. The finished textiles were then shipped back to Asia, flooding local markets and deliberately bankrupting India's domestic handloom weavers.

Q1. Categorise the drivers of the East India Company's expansion in India into Economic, Military, or Political factors using structured bullet points.

Source A **: The Corporate Directive**

"We must remind our officers that the primary purpose of our presence in India is the secure maintenance of trade. Any action by our military commanders that causes unnecessary alarm or religious outrage among the native troops risks disrupting our merchant traffic and damaging the financial returns of our shareholders in London." — Directors of the East India Company, April 1857

Corporate greed eventually triggered total collapse. The East India Company's rule was defined by aggressive economic extraction, cultural insensitivity, and racial discrimination. In May 1857, this systemic oppression triggered the Indian Rebellion (historically known as the Sepoy Mutiny). Indian soldiers within the Company's army turned on their British officers, sparking a bloody, year-long conflict across northern India. The British state intervened to brutally crush the uprising with horrific violence, but the catastrophe permanently shattered the EIC's credibility. Parliament realized that a private trading company could no longer be trusted to govern a subcontinent safely. In 1858, the Government of India Act formally abolished the East India Company and transferred total administrative control to the British Crown, establishing the British Raj.

Q2. Analyze the motive behind the EIC directive provided in the source. How does this corporate motive impact its usefulness for a historian investigating the causes of the 1857 Rebellion?

Direct rule over distant millions required absolute control of the world's shipping lanes, a strategy known as naval supremacy. Following the Napoleonic Wars, the Royal Navy enforced the strict 'Two-Power Standard' to protect merchant networks and project imperial power. This global supremacy was physically manufactured and sustained right here in Hampshire, at the Portsmouth Dockyard. Throughout the 19th century, Portsmouth was transformed into the largest steam-powered industrial complex on earth. It was here that Britain abandoned wooden sailing ships and turned to steam-driven 'Ironclads'. These floating fortresses required massive technological synthesis: they utilized the advanced metallurgy pioneered by Henry Cort at Funtley (Lesson 1) for their armor plating, and were built using the very 'Fareham Red' bricks studied in Lesson 3 for the expanding dockyard basins. Thousands of local Hampshire tradesmen—shipwrights, smiths, and boiler-makers—worked grueling shifts to build the fleets that patrolled the world.

Q3. Explain the significance of Portsmouth Dockyard in maintaining Britain's global imperial supremacy during the 19th century.

Source B **: The Riveter's Letter**

"The noise inside these iron hulls is enough to break a man's spirit. We swing heavy sledgehammers for twelve hours straight in the blistering heat of the slips, breathing the black smoke of the forges. Our skin is scarred by flying scales of red-hot iron. The masters talk of the glory of the fleet, but we know every plate on this Ironclad is paid for with the sweat and broken health of Hampshire men." — Arthur Vance, Portsmouth Dockyard Riveter, November 1861

While politicians in London championed the Navy as a glorious symbol of national pride, the reality for the industrial workforce in Hampshire was defined by exhausting toil and hazardous working conditions. The transition to iron-hulled engineering required a completely different class of hard manual labor, stripping away the traditional skills of wood craftsmen and replacing them with the raw, deafening power of the industrial forge. Historians must analyze working-class records to balance the grand strategic narratives of naval historians.

Q4. Compare the perspective of the dockyard worker in the source with the official government policy of the 'Two-Power Standard'. How does the provenance of this letter affect its reliability for a historian studying the human cost of empire?

Source C: The Weavers' Petition

> *"Our trade is entirely ruined. The cloth of England has flooded our markets, sold at a price we cannot compete with. For generations, our families have lived in comfort by our looms, but now our looms sit silent, and our children are reduced to starvation. We beg the government to place a duty on the clothing of England, or we shall perish utterly." — Weavers of the Dacca District, India, 1830*

To truly evaluate how the empire was sustained, historians must look beyond the dockyards of Hampshire and examine the devastating economic impact on the colonized populations. The British industrial economy required a continuous supply of cheap raw materials and an uncompetitive market to buy its factory outputs. This system altered the traditional lifestyles of millions. In India, local artisans who had spent generations producing world-class hand-woven textiles found themselves completely crushed by the aggressive influx of cheap, machine-made cotton goods produced in the industrial mills of northern England.

Q5. Synthesis Assessment (8 Marks): 'The British Empire was built and sustained by technological progress rather than violence and exploitation.' To what extent do you agree with this interpretation? Use the IDEA framework and evidence from across the unit (including Henry Cort, factory work, and Portsmouth Dockyard) to support your argument.

L5: How did ordinary people fight for a voice? (Peterloo, Chartism, Trade Unions; Case Study: Hampshire Swing Riots)

LEARNING OBJECTIVES

- [ ] Understand the core themes of How did ordinary people fight for a voice? (Peterloo, Chartism, Trade Unions; Case Study: Hampshire Swing Riots).

Do Now Activity

1. **Recall question from previous lesson.**
2. **What was the significance of the Battle of Hastings in 1066?**
3. **What was the Black Death?**
4. **Who was Guy Fawkes?**
5. **What was the Renaissance?**

Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Chartism | Trade Unions

Your Sentence:

Source A: The Swing Riots

Source

An illustration of agricultural workers protesting against the introduction of threshing machines during the Swing Riots.

Placeholder narrative.

Q1. Placeholder task.

L6: How did the road to democracy expand? (Reform Acts 1832–1884, Secret Ballot; Case Study: Hampshire MPs & Pocket Boroughs)

LEARNING OBJECTIVES

- [ ] Understand the core themes of How did the road to democracy expand? (Reform Acts 1832–1884, Secret Ballot; Case Study: Hampshire MPs & Pocket Boroughs).

Do Now Activity

1. Recall question from previous lesson.

2. What was the significance of the Magna Carta signed in 1215?

3. Why did Henry VIII break away from the Catholic Church?

4. What was the primary cause of the English Civil War?

5. What was the result of the Spanish Armada in 1588?

Vocabulary Check

*Write a clear definition and a historically accurate sentence for the term: **Reform Acts***

Source A: A Pocket Borough

Source

A political cartoon mocking the corruption of un-reformed electoral constituencies.

Placeholder narrative.

Q1. Placeholder task.

L7: Who party truly benefited from 19th-century transformation? (Synthesis essay balancing economic growth vs. exploitation)

LEARNING OBJECTIVES

- [ ] Understand the core themes of Who party truly benefited from 19th-century transformation? (Synthesis essay balancing economic growth vs. exploitation).

Do Now Activity

1. Recall question from previous lesson.

2. What was the significance of the Battle of Hastings in 1066?

3. What was the Black Death?

4. Who was Guy Fawkes?

5. What was the Renaissance?

Vocabulary Check

Write a short paragraph using at least FOUR of the vocabulary words below correctly.

Synthesis | Exploitation

Source A: **The Two Nations**

Source

An illustration showing the stark divide between the rich industrialists and the poor working classes in Victorian Britain.

Placeholder narrative.

Q1. Placeholder task.